

Evaluation, system comparison

Inconsistency Checker · v0.8.8 · 2026-08-18 · fixed seed 7

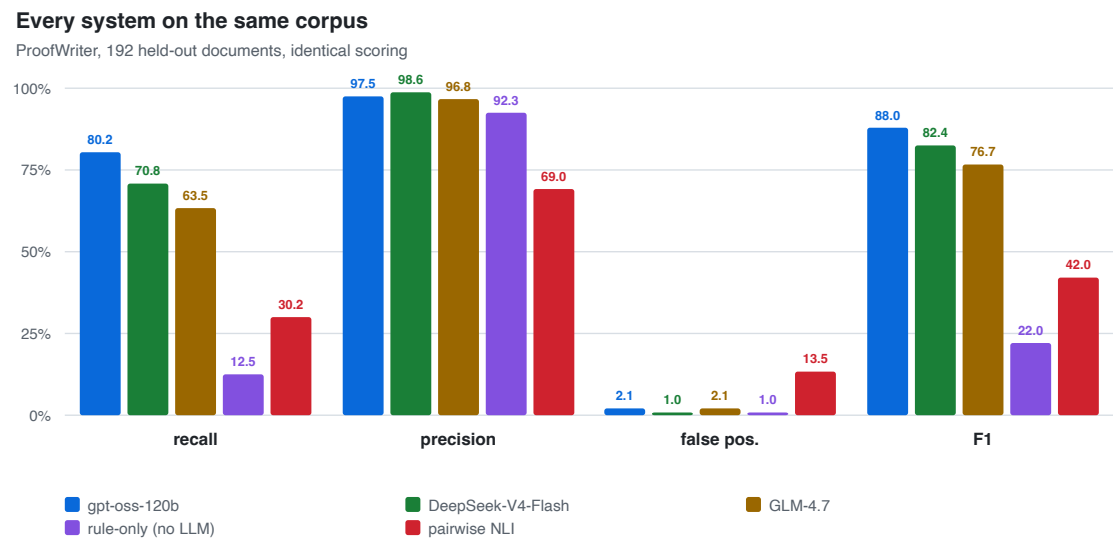
Companion to [evaluation.md](#), which reports the headline results. This page compares every system that was run, on every corpus, with all four detection metrics and the raw confusion counts behind them.

Five systems are compared. Three are the same pipeline with a different model doing the translation step; one is the pipeline with that step removed entirely; one is an external baseline.

System	What it is
gpt-oss-120b	the pipeline, default translation model
DeepSeek-V4-Flash	the pipeline, translation swapped
GLM-4.7	the pipeline, translation swapped
rule-only (no LLM)	the pipeline with the language model withheld, deterministic translation only
pairwise NLI	sentence-pair entailment: every pair judged in isolation, flag if any pair conflicts

Every number below is read from the stored run metrics by `tools/compare.py`, so the tables cannot drift from the runs that produced them. Regenerate with `python -m tools.compare`.

All systems, one corpus



ProofWriter is the only corpus on which all five ran, so it carries the comparison. Recall separates them; **precision does not**. Four of the five sit above 92% precision, and the one that does not is the baseline.

Full results

ProofWriter (n=192)

System	Recall (95% CI)	Precision	False pos.	F1	TP/FP/FN/TN	Cost/doc
gpt-oss-120b	80.2% [71.1-86.9]	97.5%	2.1%	88.0%	77/2/19/94	5,168 tok
DeepSeek-V4-Flash	70.8% [61.1-79.0]	98.6%	1.0%	82.4%	68/1/28/95	4,924 tok
GLM-4.7	63.5% [53.6-72.5]	96.8%	2.1%	76.7%	61/2/35/94	18,381 tok
rule-only (no LLM)	12.5% [7.3-20.6]	92.3%	1.0%	22.0%	12/1/84/95	offline
pairwise NLI	30.2% [21.9-40.0]	69.0%	13.5%	42.0%	29/13/67/83	103.2 calls

FOLIO (n=141)

System	Recall (95% CI)	Precision	False pos.	F1	TP/FP/FN/TN	Cost/doc
gpt-oss-120b	50.0% [38.7-61.3]	92.3%	4.3%	64.9%	36/3/36/66	4,269 tok
pairwise NLI	45.8% [34.8-57.3]	71.7%	18.8%	55.9%	33/13/39/56	16.7 calls

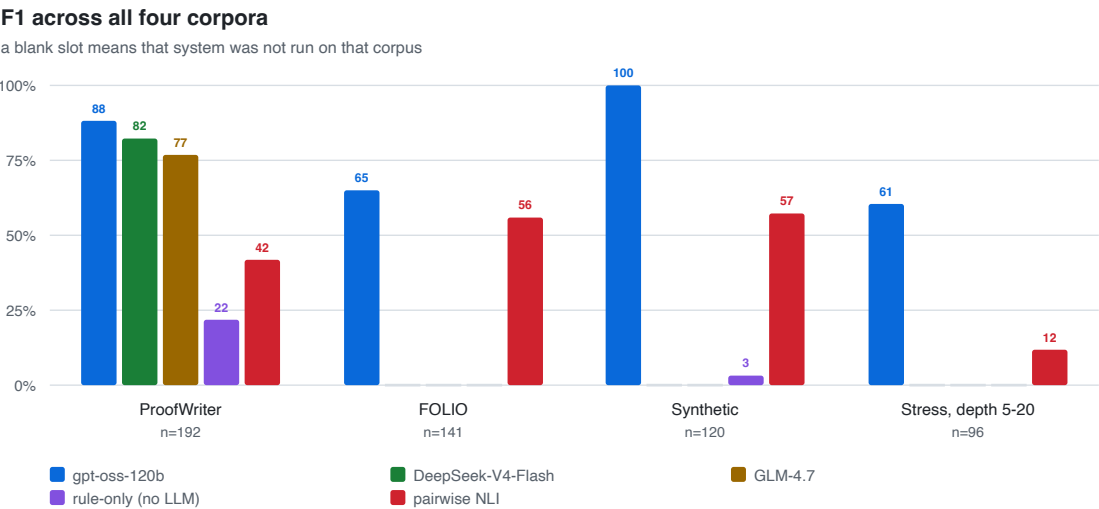
Synthetic (n=120)

System	Recall (95% CI)	Precision	False pos.	F1	TP/FP/FN/TN	Cost/doc
gpt-oss-120b	100% [94.0-100.0]	100%	0.0%	100%	60/0/0/60	3,045 tok
rule-only (no LLM)	1.7% [0.3-8.9]	100%	0.0%	3.3%	1/0/59/60	offline
pairwise NLI	40.0% [28.6-52.6]	100%	0.0%	57.1%	24/0/36/60	21.9 calls

Stress, depth 5 to 20 (n=96)

System	Recall (95% CI)	Precision	False pos.	F1	TP/FP/FN/TN	Cost/doc
gpt-oss-120b	47.9% [34.5-61.7]	82.1%	10.4%	60.5%	23/5/25/43	21,556 tok
pairwise NLI	6.2% [1.1-28.3]	100%	0.0%	11.8%	1/0/15/16	127.2 calls

F1 across corpora



A flat line at the axis marks a system that was not run on that corpus, not a score of zero. **DeepSeek and GLM were only run on ProofWriter**, so the model comparison rests on that corpus alone and should not be

generalised to the others without running them.

The pipeline leads on F1 on all four corpora. The margin is widest where the reasoning is deepest (60.5 against 11.8 on the stress set) and narrowest on FOLIO (64.9 against 55.9), whose contradictions are shallower and more lexical.

The trade-off each system makes

Precision against recall

ProofWriter; up and to the right is better, and the corner is empty



the pipeline trades recall for precision by refusing what it cannot represent; the pairwise baseline gives up both

The systems do not sit on a single quality axis; they choose different trade-offs, and the shape of that choice matters more than the ranking.

The pipeline buys precision by refusing. Content it cannot represent is quarantined with a written reason instead of guessed at, which costs recall and protects precision. Rule-only is the same policy at its limit: 92.3% precision on 12.5% recall, because the little it does translate, it translates soundly.

The baseline gets neither. 30.2% recall at 69.0% precision, with a **13.5% false-positive rate**: it flags roughly one clean document in seven. That figure also explains why its recall does not fall to zero at depth. A method that flags one in seven clean documents will sometimes flag a contradictory one for the wrong pair and be scored correct, so part of its deep-chain recall is a lucky guess rather than a found chain.

Cost runs the other way from accuracy. The baseline is the most expensive per document, at 103 model calls against a single pass, because it grows with the square of document length. On the stress set that reaches 127 calls per document for 6.2% recall.

Reading the model comparison carefully

The three translation models differ by 17 points of recall, but their intervals overlap: gpt-oss [71.1-86.9] against GLM [53.6-72.5] barely separate, and gpt-oss against DeepSeek not at all. Treat the ordering as suggestive.

They were also **not run under identical conditions**. gpt-oss cannot disable internal reasoning on its provider; DeepSeek can. GLM emitted 18,381 tokens per document against gpt-oss's 5,168 for the worst result of the three, so it is slower and more expensive as well as less accurate.

What the comparison does establish is the size of the language model's contribution, and it is large: withholding it drops recall from 80.2% to 12.5% on ProofWriter and from 100% to 1.7% on the synthetic set, while precision barely moves. **Recall is bounded by translation coverage**, not by the reasoning engine, and that is where further work belongs.

Scope: this page compares the pipeline against sentence-pair entailment. Prompting a language model to judge a whole document at once is a different approach that does not face the pairwise structural limit, and is not evaluated here or in [evaluation.md](#).